



Module No : Module – 05

Assignment No: Assignment 05

Module Title: Problem Solving with AI (Claude Code +
JavaScript)

Submitted By

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Q1. JavaScript Coding Practice.

Write Programs for the following

1. Convert Celsius to Fahrenheit

Problem Statement: Write a function that takes a temperature in Celsius and converts it to Fahrenheit

Source Code:

```
function celsiusToFahrenheit (celsius){  
    let fahrenheit = (celsius * 9 / 5 ) + 32;  
    return fahrenheit;  
}  
let celsius = 32;  
let result = celsiusToFahrenheit(32);  
console.log("Celsius: ",celsius);  
console.log("Rahrenheit: ",result);
```

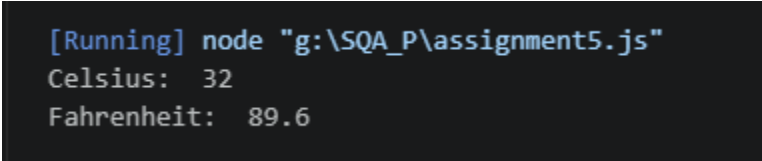
Input:

Celsius = 32

Output:

Celsius: 32

Rahrenheit: 89.6



```
[Running] node "g:\SQA_P\assignment5.js"  
Celsius: 32  
Fahrenheit: 89.6
```

Explanation: The function receives a temperature in Celsius. It applies the given conversion formula and returns the temperature in Fahrenheit. In this example, 32°C is converted into 89.6° F.

2. Find the Factorial of a Number.

Problem Statement: Write a function that calculates the factorial of a given non-negative integer.

Source Code:

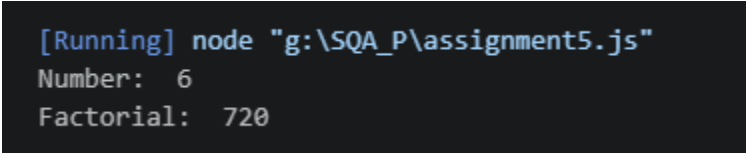
```
function factorial(number){  
  let result = 1;  
  for (let i = 1; i <= number; i++){  
    result = result * i;  
  }  
  return result;  
}  
let number = 6 ;  
let answer = factorial(number);  
console.log("Number: ", number );  
console.log("Factorial: ",answer);
```

Input:

Number: 6

Output:

Factorial: 720



```
[Running] node "g:\SQA_P\assignment5.js"  
Number: 6  
Factorial: 720
```

Explanation: The function starts with result = 1. Then a loop multiplies the result by every number from 1 to the given number. For input 6, the calculation becomes $1 * 2 * 3 * 4 * 5 * 6$, which gives 720

3. Check for Palindrome

Problem Statement: Write a function that checks whether a word reads the same backwards as forward. The function should return true or false

Source Code:

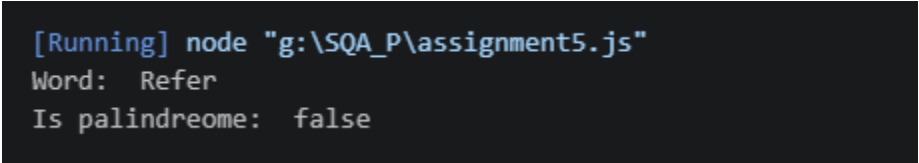
```
function isPlaindrome(word){  
  let reverse = "";  
  for(let i = word.length - 1; i >= 0; i- -){  
    reverse = reverse + word[i];  
  }  
  Return word == reverse;  
}  
let word = "Refer";  
let result = isPlaindrome(word);  
console.log("word: ",word);  
console.log("Is Palindrome: ",result);
```

Input:

Word: Refer

Output:

Is Palindrome: True



```
[Running] node "g:\SQA_P\assignment5.js"  
Word: Refer  
Is palindreome: false
```

Explanation: The function creates a reverse version of the given word. Then, it compares the original word with the reversed word. If both are the same, it returns true; otherwise, it returns false

4. Sum of Array Elements

Problem Statement: Write a function that takes an array of numbers and returns the sum of all elements.

Source Code:

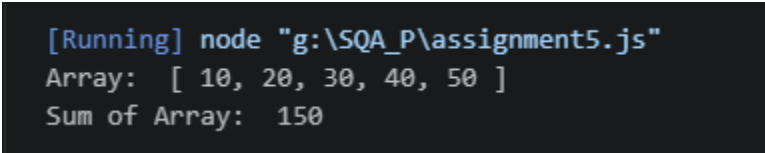
```
function sumOfarray(numbers){
    let sum = 0;
    for (let i = 0; i < numbers.length; i - -){
        Sum = sum + numbers[i];
    }
    return sum;
}
let numbers = [10, 20, 30, 40, 50];
let result = suOfarray(numbers);
console.log("Array: ",numbers);
console.log("Sum of Array: ",result);
```

Input:

Array: [10, 20, 30, 40, 50];

Output:

Sum of Array: 150



```
[Running] node "g:\SQA_P\assignment5.js"
Array:  [ 10, 20, 30, 40, 50 ]
Sum of Array:  150
```

Explanation: The function starts the sum variable at zero. It visits each element of the array and adds that value to sum. After all elements are processed, the total is returned.

5. FizzBuzz

Problem Statement :

Source Code:

```
for(let i = 1 ; i <=15; i ++){  
    if( i % 3 == 0 && i % 5 == 0){  
        console.log("FizzBuzz");  
    }  
    else if( i % 3 == 0 ){  
        console.log("Fizz");  
    }  
    else if( i % 5 == 0 ){  
        console.log("Buzz");  
    }  
    else{  
        console.log(i);  
    }  
}
```

Input: Number From 1 to 15

Output: 1

2

Fizz

4

Buzz

Fizz

7

8

Fizz

Buzz

11

Fizz

13

14

FizzBuzz

```
[Running] node "g:\SQA_P\assignment5.js"  
1  
2  
Fizz  
4  
Buzz  
Fizz  
7  
8  
Fizz  
Buzz  
11  
Fizz  
13  
14  
Fizz Buzz
```

Explanation: The loop runs from 1 to 15. For each number, the program checks divisibility. The condition for both 3 and 5 is checked first, so 15 prints "FizzBuzz" instead of only "Fizz" or "Buzz". Otherwise, numbers are printed according to the given rules